



scDFM

Distributional Flow Matching for robust Single-Cell Perturbation Prediction

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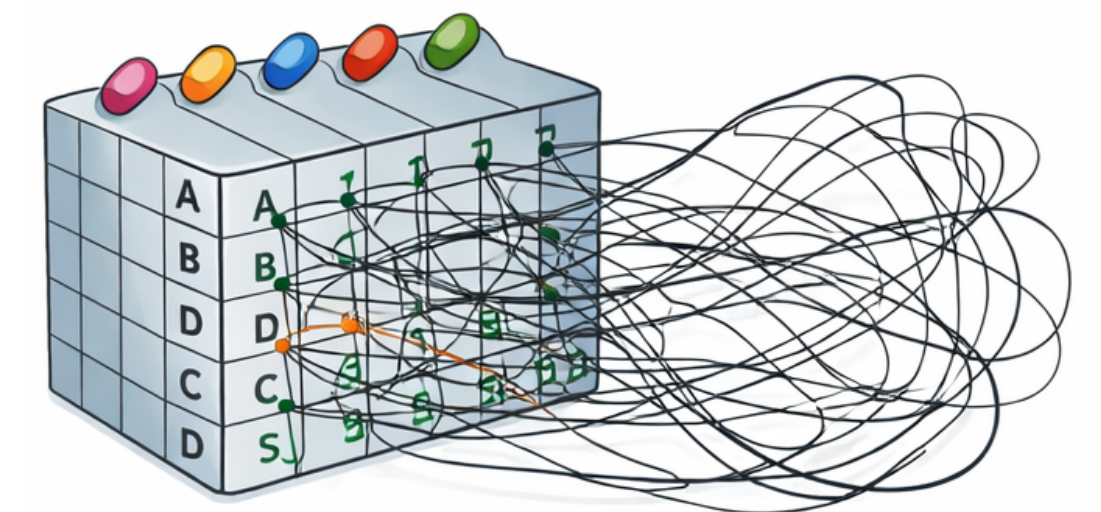
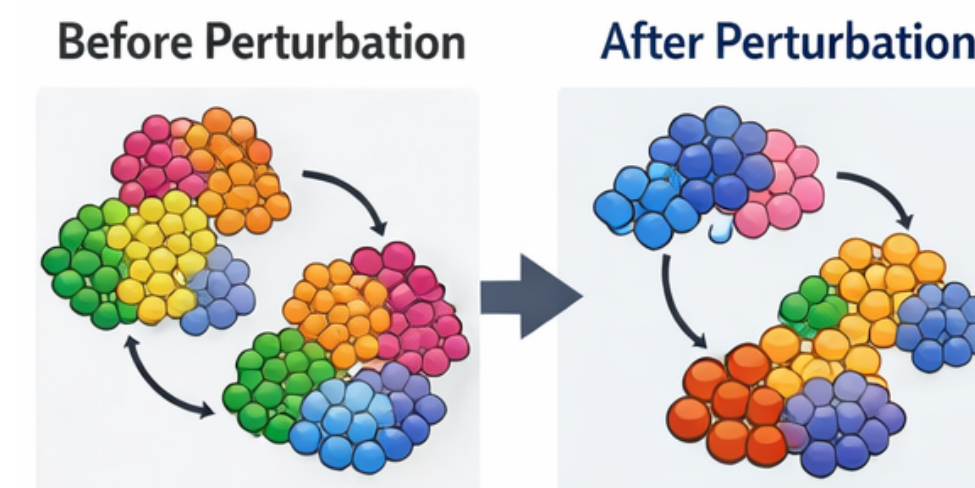
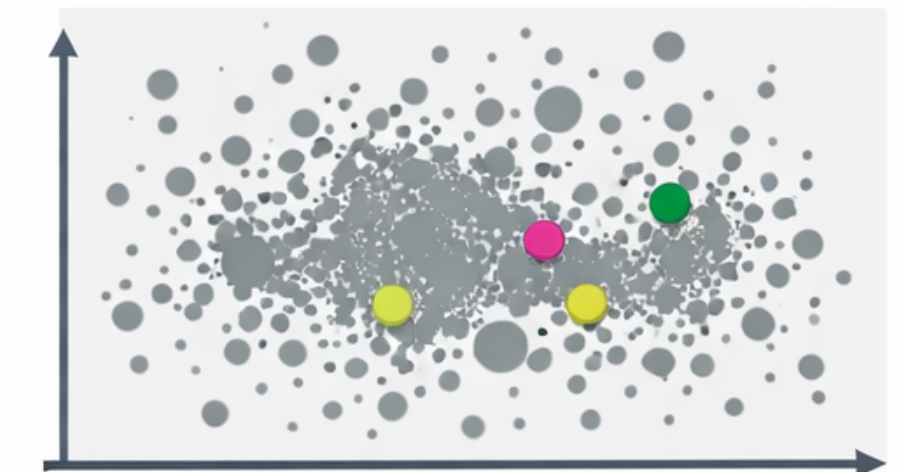
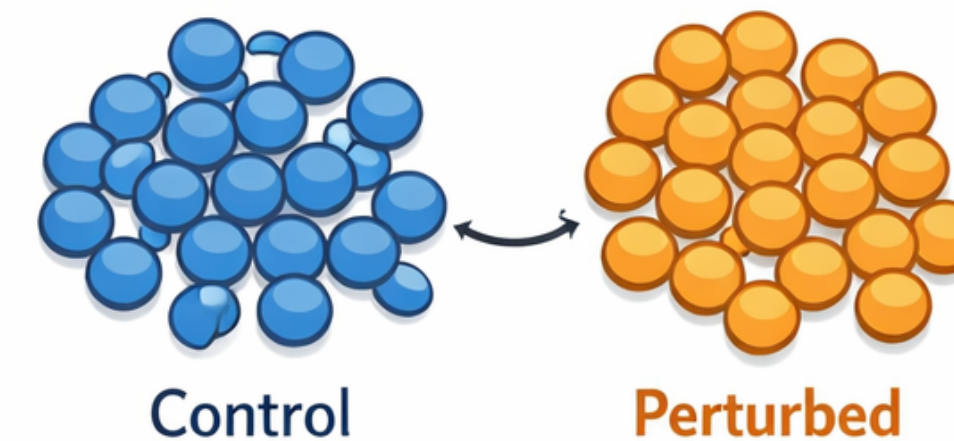
Motivation

Single-cell perturbation prediction is a distribution shift problem



Why is this hard?

- Unpaired data
- High noise & Sparsity
- Perturbations induce population-level shifts
- Combinatorial space explodes for multi-gene/Multi-drug perturbation



Motivation

Single-cell perturbation prediction is a distribution shift problem

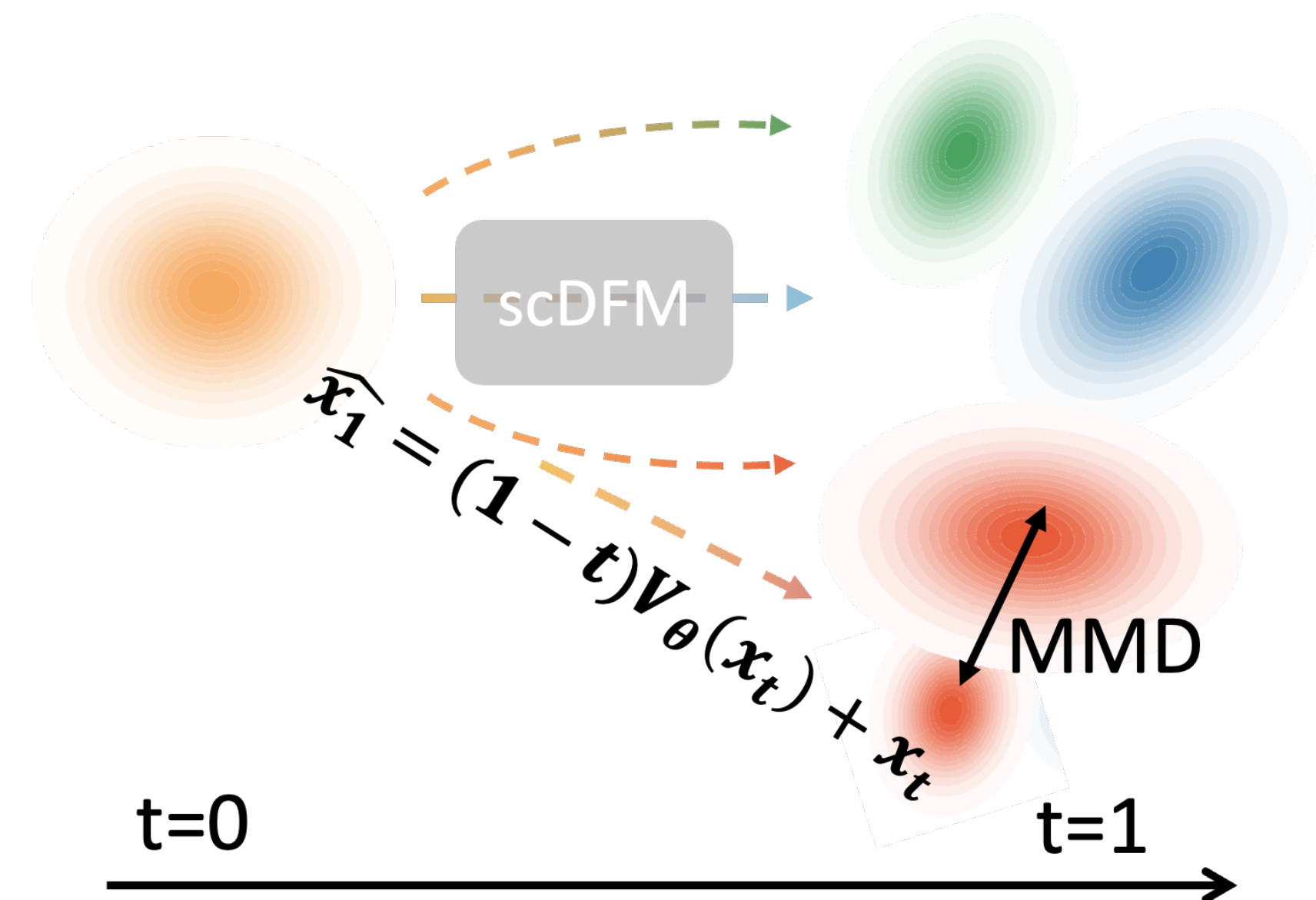


Common pitfall

Assuming cell-level correspondence
(Control Cell $\leftrightarrow$ Perturbed Cell)

What we want instead

Model the whole
perturbed population



Key Idea

From “cell matching” to conditional distribution generation



Reformulate perturbation prediction as :

Learn $p_{\theta}(x | c_x, c_p)$ that generates the entire perturbed population

c_x = control state, c_p =perturbation condition

Flow Matching

Learn a continuous velocity field to transform samples from source $\rightarrow$ target distributions.

MMD Alignment

Regularize endpoint to match the true perturbed population (multi-kernel MMD).

PAD-Transformer

Noise-robust backbone using gene-graph masks + differential attention.

Method

Conditional Flow Matching + distribution-level regularization

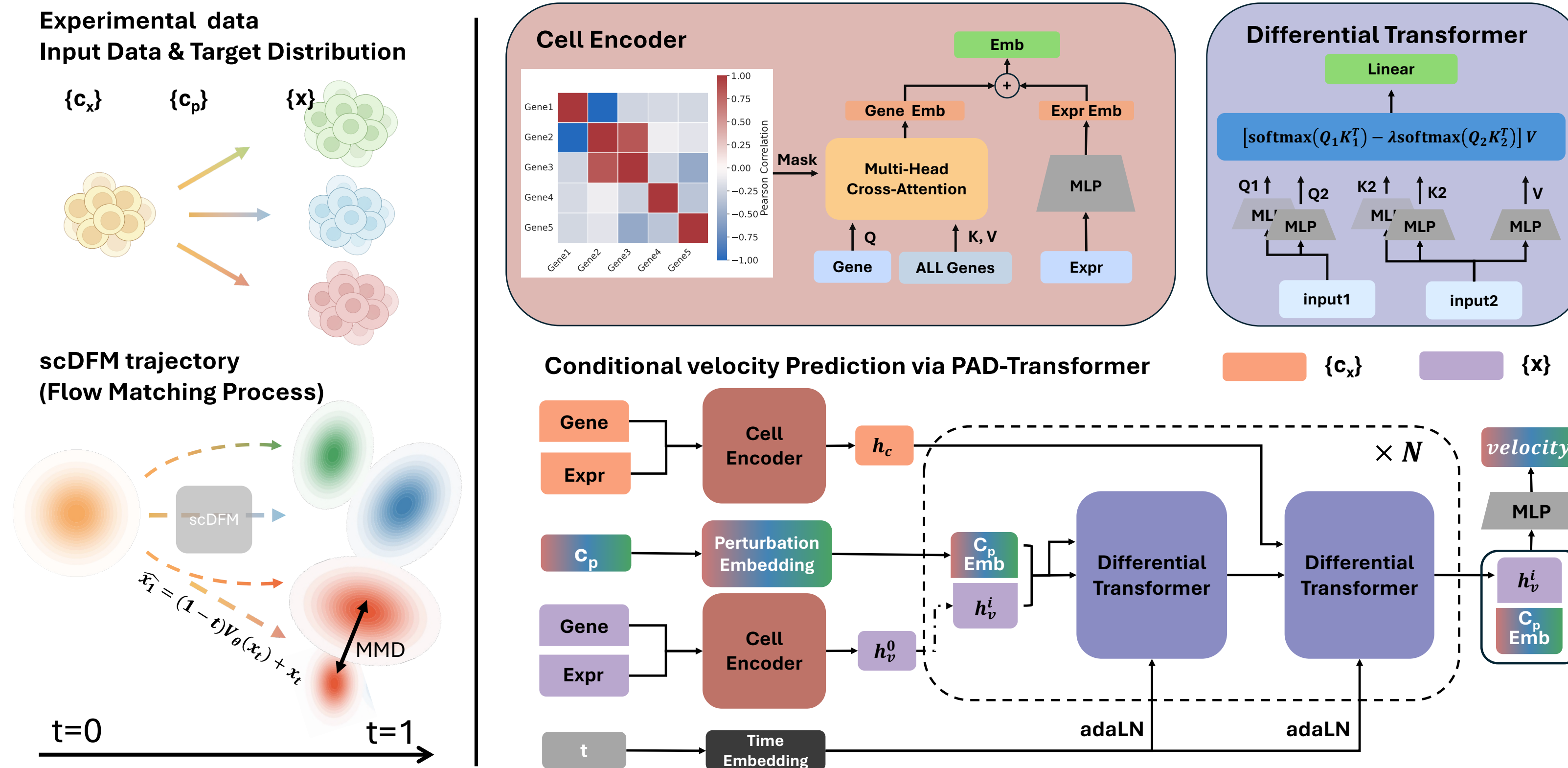


Figure: scDFM overview (flow matching trajectory + PAD-Transformer velocity predictor + MMD)

Training objective: $L = L_{CFM} + \lambda L_{MMD}$

Method



Conditional Flow Matching + distribution-level regularization

1. Learns a time-dependent velocity field that continuously transforms control cells into perturbed cells.

$$L_{\text{CFM}}(\theta) = \mathbb{E}_{c_x, c_p} \mathbb{E}_{x_0 \sim q_0, x_1 \sim q_1(\cdot | c_x, c_p)} \mathbb{E}_{t \sim \mathcal{U}(0,1)} \left[\left\| v_{\theta}(x_t | t, c_x, c_p) - v(x_t | x_0, x_1, t, c_x, c_p) \right\|_2^2 \right].$$

2. Multi-kernel Gaussian RBF loss enforces distributional fidelity between predicted and real cell populations.

$$L_{\text{MMD}}(\theta) = \mathbb{E}_{c_x, c_p} \mathbb{E}_{x_0 \sim q_0, x_1 \sim q_1(\cdot | c_x, c_p)} \mathbb{E}_{t \sim \mathcal{U}(0,1)} k_{\text{mix}} \left(x_1, \hat{x}_1(x_t, t, c_x, c_p) \right)$$

3. Uses perturbation-aware differential attention and gene-graph context to predict biologically meaningful velocity fields.

$$L = L_{\text{CFM}} + \lambda L_{\text{MMD}}$$

Experimental setup

Datasets, splits, and metrics



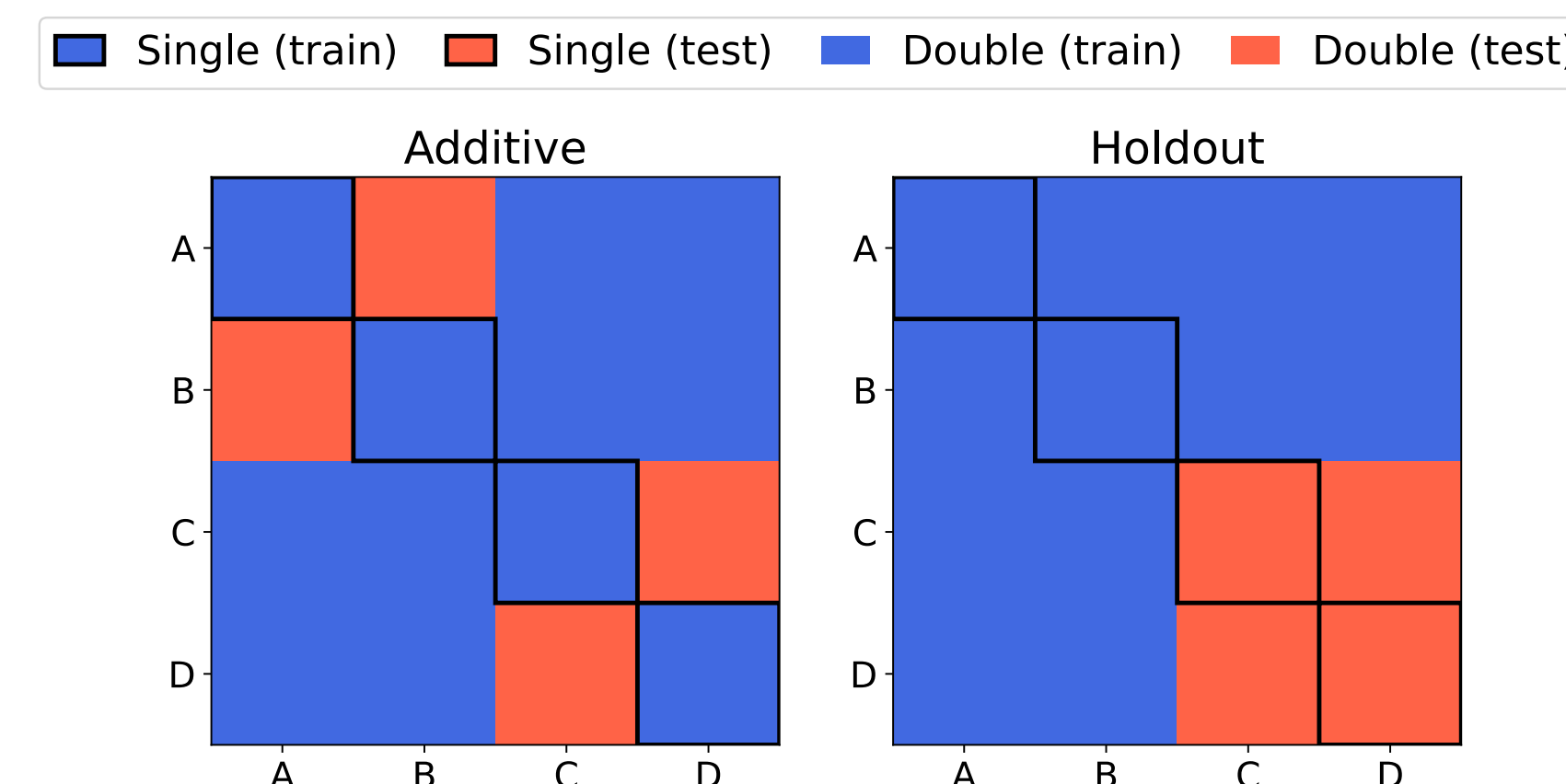
Benchmarks

- Norman (CRISPR genetic perturbations; single + double)
- ComboSciPlex (combinatorial drug perturbations)

Evaluation splits

- Additive: all singles seen; test unseen doubles made from seen singles
- Holdout: remove some singles and all doubles involving them → test unseen singles & doubles

Split schematic

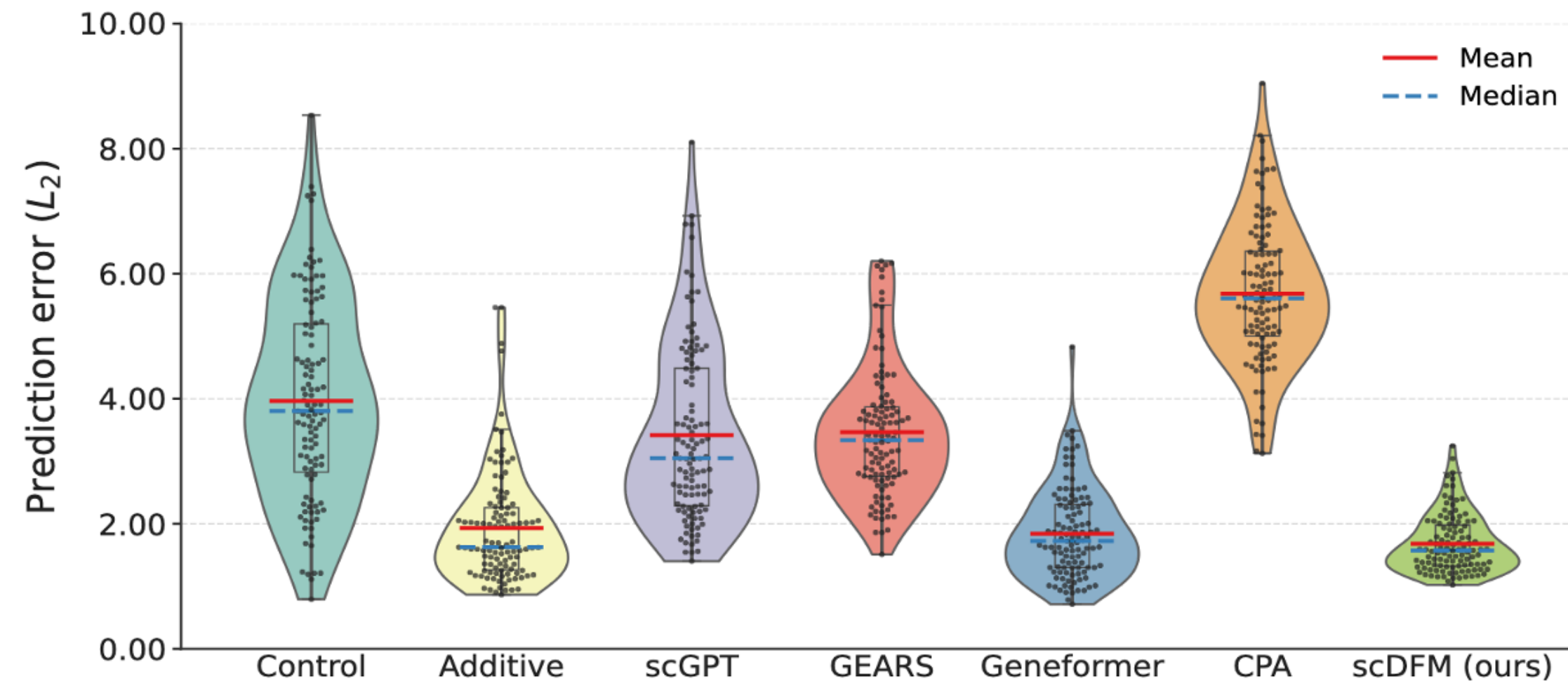


Metrics

1. Pointwise: MAE / MSE / L2, Pearson Δ
2. Distribution: DS (discrimination score)
3. Differential expression: DE-Spearman (rank corr. on DE genes)

Results

Norman(Additive)



MSE vs best baseline

-19.6%

0.00315 vs 0.00392 (CellFlow)

DS

0.9737

Higher is better

DE-Spearman

0.5705

Captures DE gene ranking

Table 1: Comparison of different methods across evaluation metrics on the Norman additive split.

Model	L2 ↓	MSE ↓	MAE ↓	DE-Spearman ρ ↑	Pearson Δ ↑	DS ↑	Pearson $\hat{\Delta}$ ↑	Pearson $\hat{\Delta}_{20}$ ↑
Control	3.9937	0.01839	0.03953	N.A.	N.A.	0.5135	-0.1695	-0.1297
Additive	1.9395	0.00448	0.02276	0.5564	0.9024	<u>0.9686</u>	0.8584	<u>0.9244</u>
scGPT	3.4112	0.01349	0.03796	1.07e-5	0.5304	0.5404	0.2165	0.2414
Geneformer	1.9132	0.00410	0.02360	0.3741	0.7732	0.8241	-0.0078	0.2239
GEARS	3.5531	0.01387	0.06624	<u>0.5624</u>	0.7421	0.8601	-0.0089	0.2032
CPA	5.7629	0.03435	0.07894	0.0713	0.3845	0.6021	-0.0039	0.2254
STATE	17.3330	0.30059	0.24705	0.5288	-0.0108	0.5135	-0.0069	0.2515
CellFlow	<u>1.7064</u>	<u>0.00392</u>	<u>0.02207</u>	0.5503	0.8678	0.9321	0.8395	0.8988
scDFM (ours)	1.7043	0.00315	0.02155	0.5705	<u>0.8853</u>	0.9737	<u>0.8468</u>	0.9260

Results

Norman(Holdout)



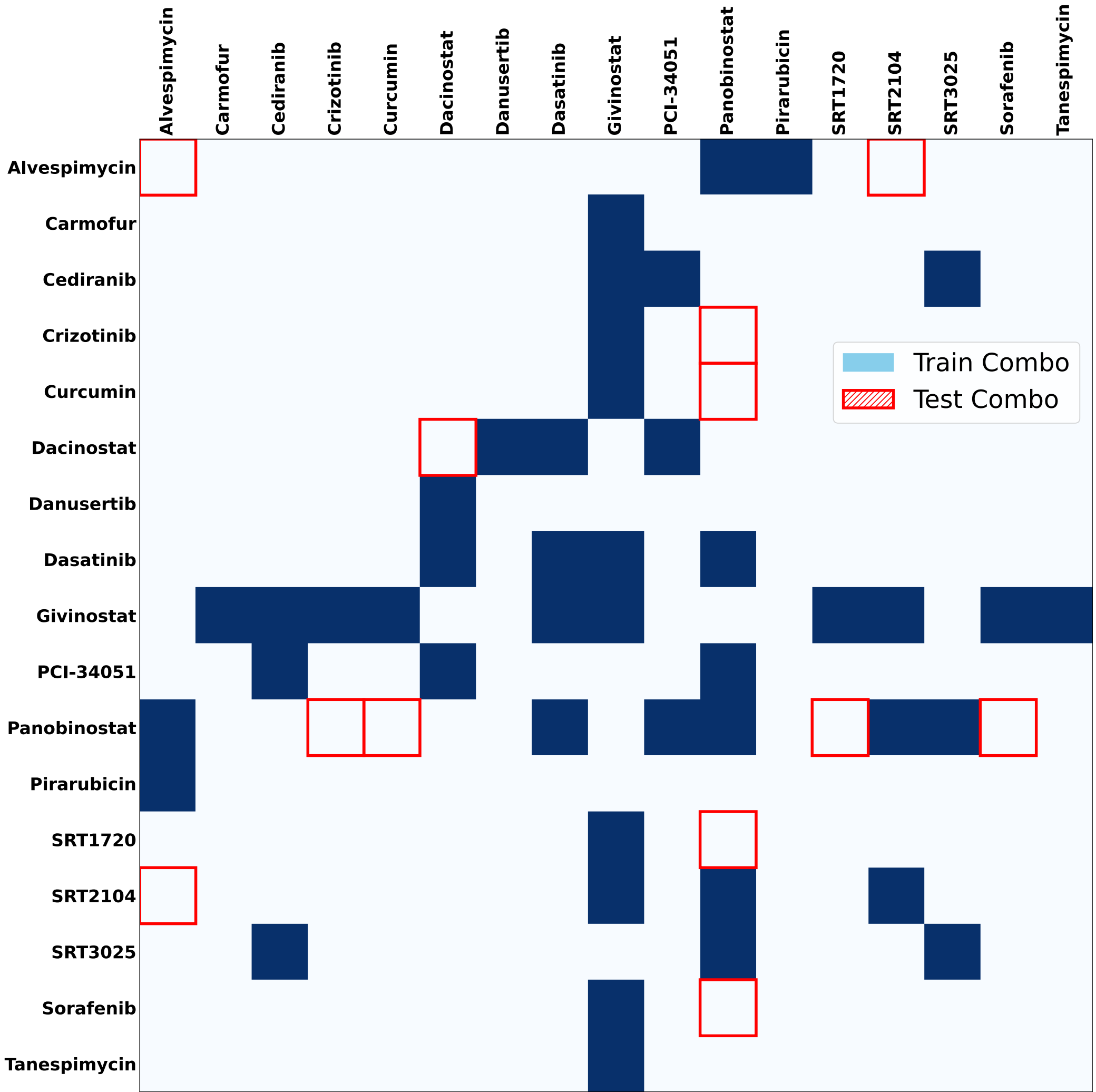
Setting	Model	L2 ↓	MSE ↓	MAE ↓	DE-Spearman ρ ↑	Pearson Δ ↑	DS ↑	Pearson $\hat{\Delta}$ ↑	Pearson $\hat{\Delta}_{20}$ ↑
Single	Control	2.6834	0.0095	0.0263	N.A.	N.A.	0.5217	0.1618	0.1982
	scGPT	2.5007	0.0080	0.0259	-0.1139	0.4503	0.5680	0.0747	0.0798
	GEARS	2.5641	0.0075	0.0466	0.3569	0.6646	<u>0.8271</u>	<u>0.6356</u>	<u>0.7914</u>
	Geneformer	1.6962	0.0036	<u>0.0191</u>	0.3669	0.6955	0.8070	0.5620	0.6513
	CPA	5.8060	0.0356	0.0853	0.1168	0.2837	0.5796	-0.0028	0.0802
	STATE	18.2543	0.3333	0.2693	<u>0.6116</u>	0.0004	0.5236	0.0154	0.2386
	CellFlow	<u>1.6758</u>	<u>0.0035</u>	<u>0.0191</u>	0.2860	<u>0.7109</u>	0.8072	0.6138	0.6753
	scDFM (ours)	1.6186	0.0030	0.0190	0.6957	0.7127	0.8914	0.6659	0.8116
Double	Control	4.1882	0.0207	0.0423	N.A.	N.A.	0.5322	-0.1303	-0.0265
	scGPT	3.5171	0.0153	0.0362	-0.0665	0.5693	0.5578	0.2814	0.2652
	GEARS	3.7458	0.0156	0.0708	0.2543	0.7552	<u>0.8766</u>	0.6407	<u>0.8413</u>
	Geneformer	<u>2.0819</u>	0.0050	0.0237	0.3468	0.7361	0.8067	0.6245	0.7261
	CPA	5.7891	0.0357	0.0796	0.3652	0.4176	0.6311	0.2432	0.2870
	STATE	18.4458	0.3404	0.2733	0.4071	0.0061	0.5289	-0.0023	0.2580
	CellFlow	2.1042	<u>0.0049</u>	<u>0.0236</u>	<u>0.5074</u>	<u>0.8095</u>	0.8622	<u>0.6780</u>	0.7155
	scDFM (ours)	2.0309	0.0047	0.0235	0.5676	0.8357	0.9189	0.7769	0.8688

Results

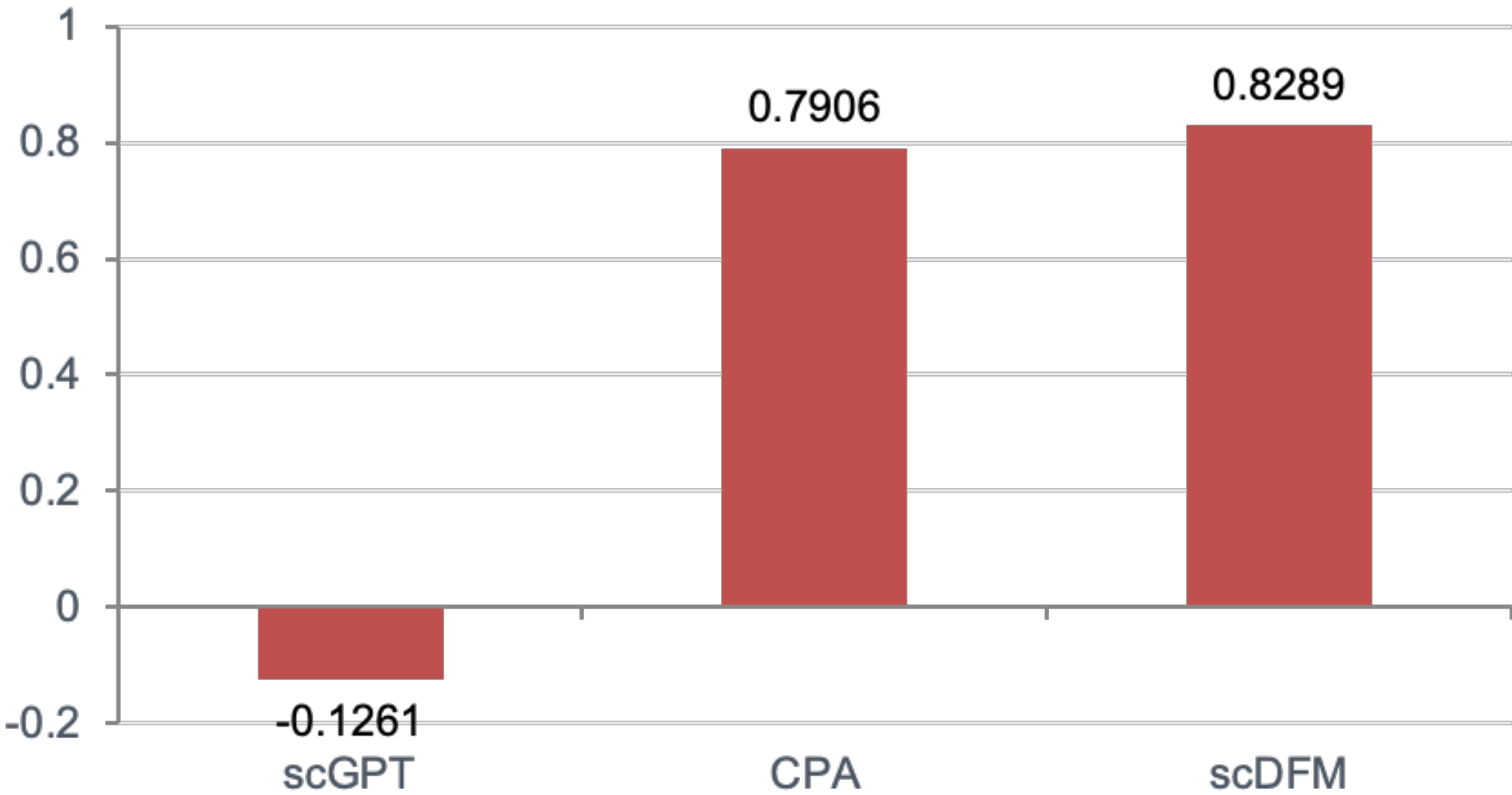
ComboSciPlex



Model	L2 ↓	MSE ↓	MAE ↓	DE-Spearman ρ ↑	Pearson Δ ↑	DS ↑
Control	5.3716	0.0324	0.0698	N.A.	N.A.	0.5714
scGPT	1.6934	0.0031	0.0251	-0.1261	<u>0.8322</u>	0.8571
CPA	<u>1.6592</u>	<u>0.0029</u>	<u>0.0240</u>	<u>0.7906</u>	0.8150	0.8980
scDFM (ours)	1.6567	0.0028	0.0220	0.8289	0.8933	<u>0.8776</u>

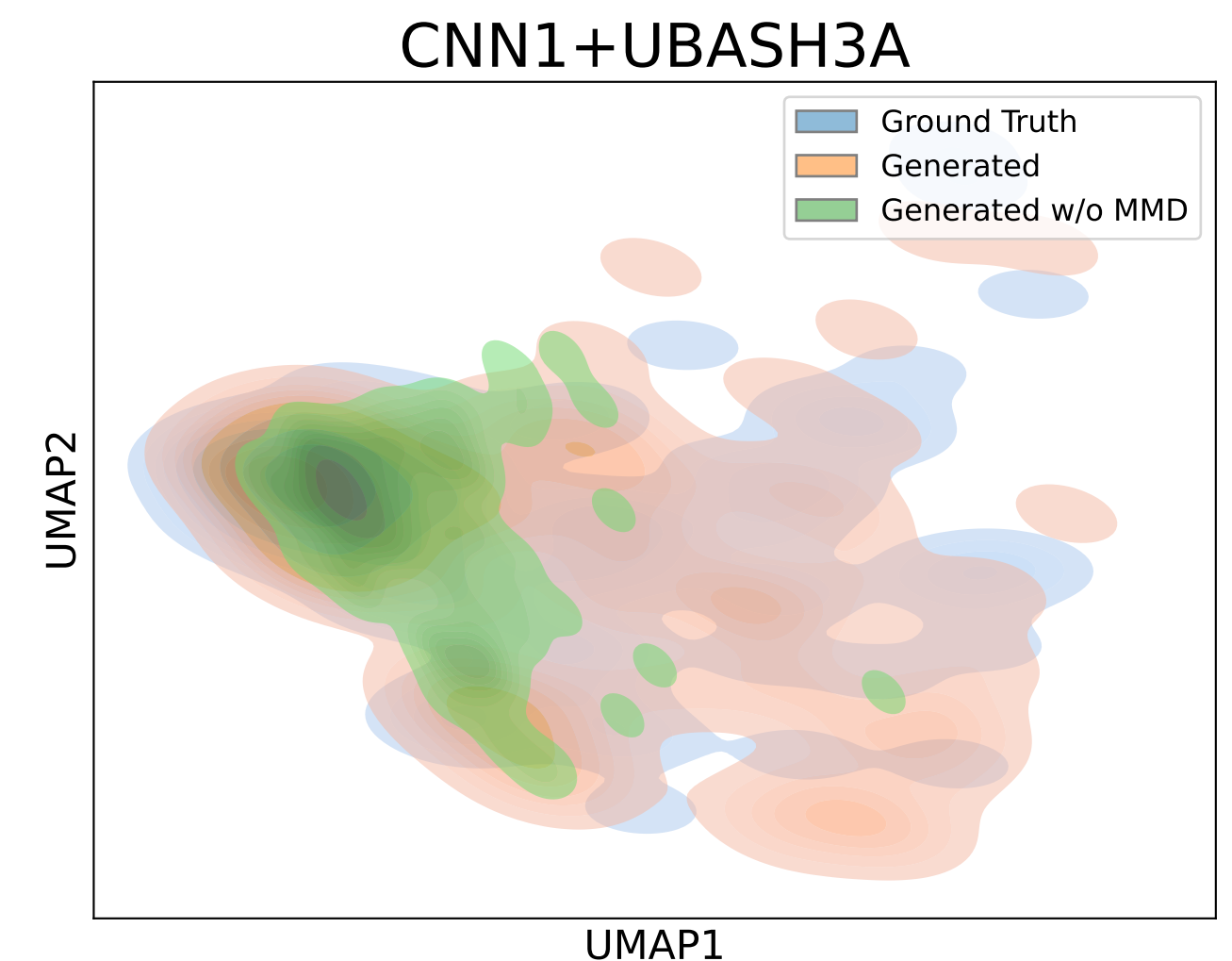
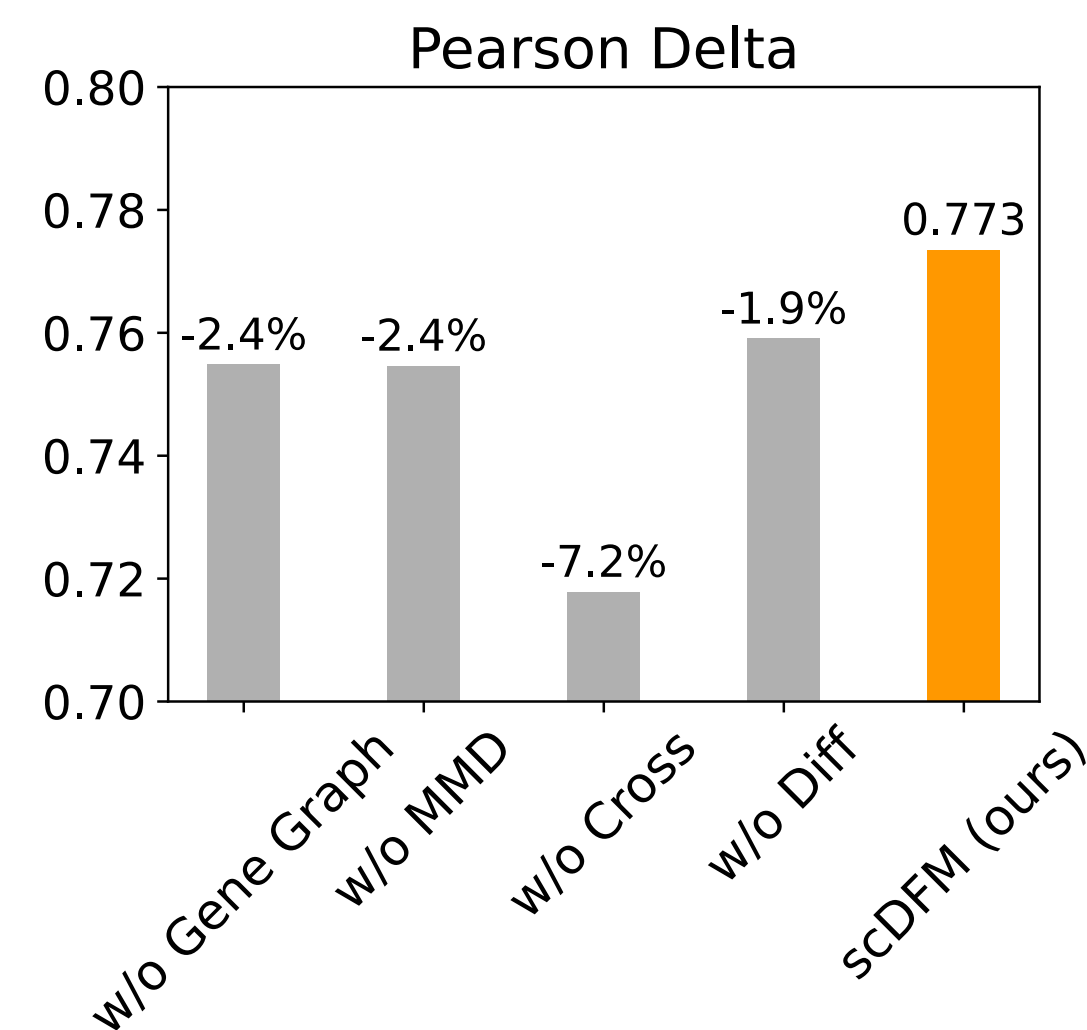
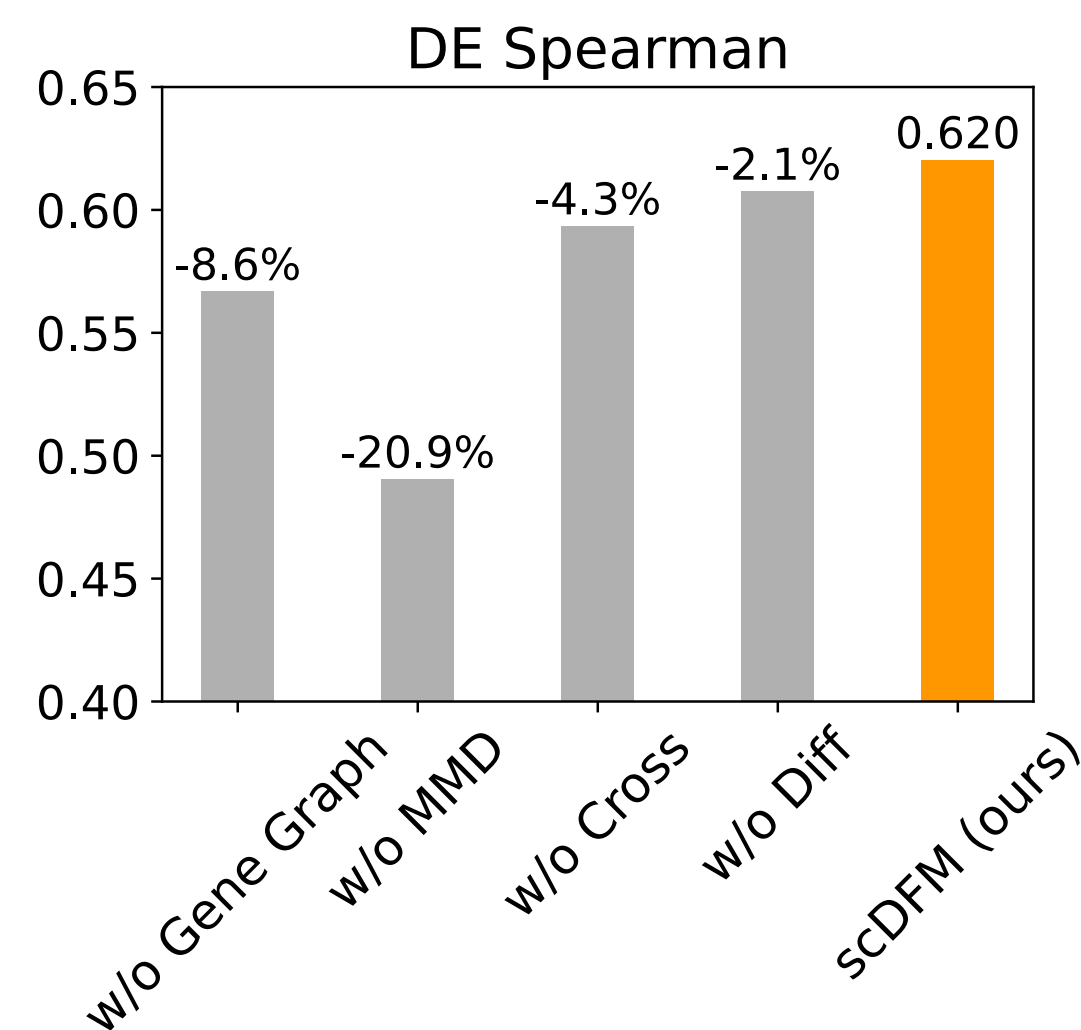
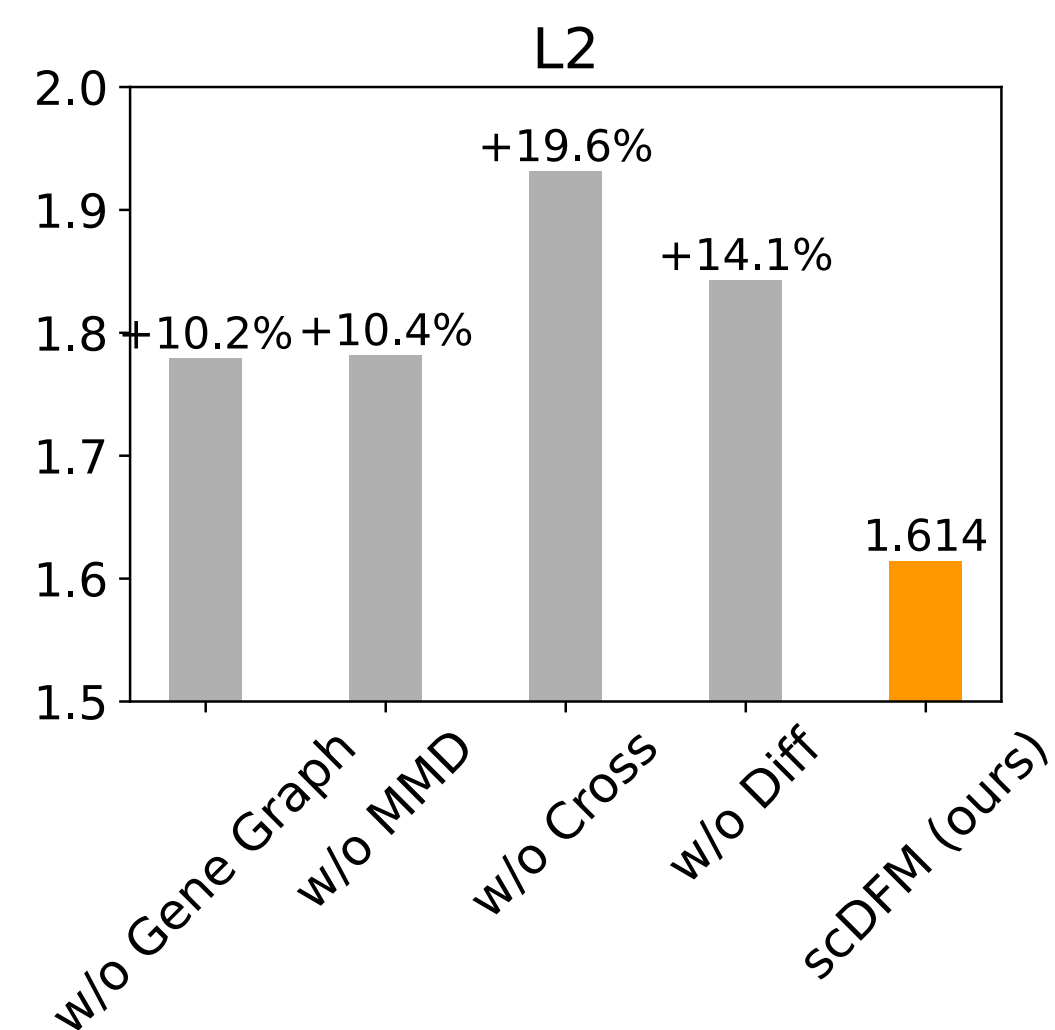


DE-Spearman on ComboSciPlex



Why does it work?

Ablations + qualitative distribution alignment



- MMD is critical for endpoint population fidelity
- Gene-graph mask improves robustness to noisy genes
- Differential attention improves DE signal recovery

Observation

MMD aligns the endpoint distribution → stable, biologically consistent predictions.

Takeaways



Distribution-level generative modeling for perturbation prediction

Main contributions

- Formulate perturbation prediction as conditional distribution generation
- Flow matching in expression space + MMD for endpoint distribution fidelity
- PAD-Transformer: gene-graph mask + differential attention for robustness to noise/sparsity
- Strong generalization to unseen and combinatorial perturbations (genes and drugs)



Outlook

Towards “digital twins” of cellular response: simulate large perturbation spaces in silico to prioritize experiments.